

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Peach Bottom Atomic Power Station, Unit 2

☒ **050**
☐ **052****2. Docket Number**

00277

3. Page**1 OF 3****4. Title**

Automatic Reactor Scram due to an Invalid Generator Lockout

5. Event Date

Month	Day	Year
01	29	2024

6. LER Number

Year	Sequential Number	Revision No.
2024	- 001 -	00

7. Report Date

Month	Day	Year
03	21	2024

8. Other Facilities Involved

Facility Name

☐ **050**

Docket Number

Facility Name

☐ **052**

Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)**10 CFR Part 20**☐ 20.2203(a)(2)(vi)**10 CFR Part 50**☐ 50.73(a)(2)(ii)(A)☐ 50.73(a)(2)(viii)(A)☐ 73.1200(a)☐ 20.2201(b)☐ 20.2203(a)(3)(i)☐ 50.36(c)(1)(i)(A)☐ 50.73(a)(2)(ii)(B)☐ 50.73(a)(2)(viii)(B)☐ 73.1200(b)☐ 20.2201(d)☐ 20.2203(a)(3)(ii)☐ 50.36(c)(1)(ii)(A)☐ 50.73(a)(2)(iii)☐ 50.73(a)(2)(ix)(A)☐ 73.1200(c)☐ 20.2203(a)(1)☐ 20.2203(a)(4)☐ 50.36(c)(2)☒ 50.73(a)(2)(iv)(A)☐ 50.73(a)(2)(x)☐ 73.1200(d)☐ 20.2203(a)(2)(i)**10 CFR Part 21**☐ 50.46(a)(3)(ii)☐ 50.73(a)(2)(v)(A)**10 CFR Part 73**☐ 73.1200(e)☐ 20.2203(a)(2)(ii)☐ 21.2(c)☐ 50.69(g)☐ 50.73(a)(2)(v)(B)☐ 73.77(a)(1)☐ 73.1200(f)☐ 20.2203(a)(2)(iii)☐ 50.73(a)(2)(i)(A)☐ 50.73(a)(2)(v)(C)☐ 73.77(a)(2)(i)☐ 73.1200(g)☐ 20.2203(a)(2)(iv)☐ 50.73(a)(2)(i)(B)☐ 50.73(a)(2)(v)(D)☐ 73.77(a)(2)(ii)☐ 73.1200(h)☐ 20.2203(a)(2)(v)☐ 50.73(a)(2)(i)(C)☐ 50.73(a)(2)(vii)☐ **OTHER** (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

Wade Scott, Regulatory Assurance Manager

Phone Number (Include area code)

7174563047

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	TB	RLY	K080	Y					

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 01/29/2024 at 1202 hours, Unit 2 experienced a Generator lockout, Turbine trip and automatic Reactor Scram. All control rods inserted, and all safety systems responded as designed. Reactor level was initially maintained by Feedwater and pressure controlled with bypass valves. During the scram response, condenser vacuum degraded due to the loss of Turbine steam seals, resulting in level and pressure control being transferred to Reactor Core Isolation Cooling (RCIC) and High Pressure Coolant Injection (HPCI), respectively.

The cause of the Generator lockout is under investigation, with the suspected cause being a spurious trip of the Audio Tone system. The system has been removed from service with a redundant system providing the protective function. The Audio Tone system is scheduled for replacement in the 2024 refueling outage.

This event is reportable under 10CFR50.73(a)(2)(iv)(A) due to manual or automatic actuation of systems listed in 10CFR50.73(a)(2)(iv)(B) including the Reactor Protection System, containment isolation signals, actuation of an emergency core cooling system and Reactor Core Isolation Cooling. An ENS notification was made within 4 hours of the event and was updated within 8 hours, reference ENS 56936.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Peach Bottom Atomic Power Station, Unit 2	☒ 050	2. DOCKET NUMBER 00277	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE**Plant Operating Conditions Before the Event**

Prior to the event, Peach Bottom Atomic Power Station, Unit 2 was operating at approximately 100% power in MODE 1. There were no structures, systems or components (SSC) that were inoperable at the start of the event that contributed to the event.

Event Description

On 01/29/2024 at 1140 hours, the Main Control Room (MCR) received a Generator Tone Trouble Alarm, which immediately reset once acknowledged. A second Generator Tone Trouble Alarm was received, which remained locked in for a short period of time (approximately 15 seconds) before resetting.

At 1202 hours, Unit 2 experienced a Generator lockout, Turbine trip and automatic Reactor Scram. Turbine control valves (EIS: TA) closed as expected. Safety relief valves (EIS: SB) were used to control the expected pressure increase and were not reopened after the initial transient. Primary Containment Group II and III Isolations (EIS: JM) occurred as expected. Reactor vessel level was initially controlled by Feedwater (EIS: SJ). Condenser vacuum slowly degraded due to an issue with the turbine steam seal system (EIS: TC) and, therefore, after approximately 20 minutes level control was transferred to the Reactor Core Isolation Cooling (RCIC) system (EIS: BN). Pressure control was maintained by turbine bypass valves (EIS: SO) and the High Pressure Coolant Injection (HPCI) system (EIS: BJ), which was only used for pressure control and not injection. Steam seals were restored and condenser vacuum was re-established. Vessel level control was transferred to Condensate (EIS: SD) and the unit was safely transitioned to a cold shutdown condition.

A review of the plant event logs confirmed that a generator lockout and power-load unbalance occurred. The 2-50F-386A, Main Transformer Over Head Line Protection Lock Out Relay was the first actuating device. The 'TTR-1' Audio Tone drawers indicated that a trip signal was transmitted from the Audio Tone drawers in the Cable Spreading Room to the drawers in the substation. The substation drawers opened the CB-225 and CB-215 breakers (EIS: EL) upon receipt of the trip signal.

At this time, the cause of the Generator lockout is still under investigation by a Root Cause analysis. The most probable cause is a failure of the Audio Tone trip system cables which are planned for abandonment in the upcoming 2024 refueling outage as part of a planned system upgrade. Currently, there is no plan to supplement this LER; however, this LER will be supplemented if the conclusions of the Root Cause analysis substantially change the information presented in this LER.

An investigation was performed and the cause of the loss of condenser vacuum is attributed to the setting of the sealing steam header pressure control valve. This resulted in the valve not opening to admit steam to the sealing steam header following the trip of the main turbine, which provided a path for air in-leakage and degraded condenser vacuum.

Unit 3 was unaffected by this event. Unit 2 start up was commenced at 2028 hours on 01/30/2024 and the unit was synchronized to the grid at 1427 hours on 01/31/2024.

This event is reportable under 10CFR50.73(a)(2)(iv)(A) due to manual or automatic actuation of systems listed in 10CFR50.73(a)(2)(iv)(B) including RPS, containment isolation signals, actuation of an emergency core

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☒ 050☐ 052**2. DOCKET NUMBER**

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YEAR	SEQUENTIAL NUMBER	REV NO.
2024	- 001	- 00

NARRATIVE

cooling system and RCIC. An ENS notification was made within 4 hours of the event and was updated within 8 hours, reference ENS 56936.

Safety Consequences

The reactor Scram occurred as designed and resulted in no safety consequences. All safety systems responded as expected. As described above, the temporary degradation in condenser vacuum resulted in use of the RCIC system to control level for a portion of the response, but this did not challenge the ability to safely shutdown and cool down the unit.

Corrective Actions

Immediate corrective actions were taken to remove the affected Audio Tone system ('TTR-1') from service and utilize the fully redundant 'TTR-2' system. The 'TTR-1' Audio Tone drawer has been replaced and a new set of cables has been identified and tested. After the completion of additional testing, this system will be maintained in a standby condition as a spare. The Audio Tone system is scheduled for a complete replacement in the upcoming 2024 refueling outage.

The corrective action to address the condenser vacuum degradation is to revise the procedure which sets the sealing steam pressure control valve to ensure it will receive the appropriate demand signal when needed.

Previous Similar Events

Peach Bottom has not experienced a Scram due to a Generator lockout in at least 10 years. The station experienced a trip of the Audio Tone system for an offsite source due to cable degradation in 2001 (Ref. EN 38241).